

FYP22 Monthly Progress Report: August 2026

Group: 27	Project: Beyond Optical PPG: mmWave Radar Arterial Pulse Sensing
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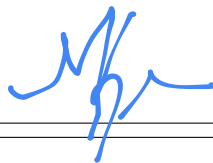
Overall Monthly Summary

August was the month our work shifted from planning into hands-on implementation and feasibility evaluation. The Infineon BGT60TR13C radar and Polar H10 chest strap arrived, and as a group we collected our first raw radar data, initially through the evaluation board's application GUI and later through the radar's Python wrapper, which gave us direct control over acquisition parameters and access to the raw ADC data rather than a processed output. We also went through the BGT60TR13C datasheet in detail to identify the allowed ranges for operating frequency, samples per chirp, chirp repetition time and frame rate, and the specific registers used to control each of these, since our adaptive algorithm idea depends on being able to change these settings on demand.

Building on the adaptive radar concept raised at the end of July, we formalised it as a contextual bandit problem, using the motion state from the IMU as context and a signal-quality-based reward to select radar chirp parameters on a per-frame basis. To compute this reward we defined our own metric, which we call CSNIR (Cardiac Signal-to-Interference-plus-Noise Ratio), and after comparing a phase-based and an amplitude-based formulation we chose to keep the reward in the amplitude/autocorrelation domain, to stay consistent with the rest of our processing pipeline and to be more robust under motion. When we tried to move this from design towards implementation, we found that the evaluation board's pre-programmed GUI does not give us the low-level, per-frame access to radar settings that a real-time adaptive loop requires, so we decided that the closed-loop adaptive algorithm will instead be implemented on our custom PCB in a later phase, while the evaluation board is used in the meantime to collect baseline pilot data under fixed settings.

Alongside this, we made progress on selecting the core hardware for the wearable prototype. We compared the Texas Instruments IWR1443 and IWR6843 against the Infineon BGT60TR13C for the radar transceiver and confirmed BGT60TR13C as our choice, and compared the MAX86176 and MAX30101 optical PPG sensors, selecting the MAX86176 for its clinical-grade design and suitability for wearables. We also began initial PCB planning, looking into wireless data transmission options, component placement and sizing constraints around the radar sensor, and the use of an FRAM for storing raw data samples. For the feasibility report we designed a block diagram of the prototype architecture and prepared a budget plan totalling approximately 644.2 USD, and prepared and revised the feasibility presentation covering the research motivation, system architecture, adaptive algorithm rationale and project timeline. We also updated the research proposal to reflect the redefined project goals, including adding blood pressure estimation as an objective, and revised the ethics documentation and participant information sheet accordingly, continuing the Sinhala and Tamil translation work ahead of submission.

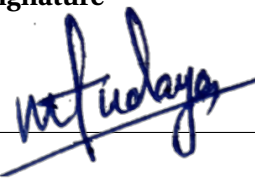
Index no.: 220029T	Name: Ananthakumar T.
Detailed list of work carried out (including problems encountered and solutions). During August, I focused on the practical implementation and feasibility evaluation of the proposed mmWave radar-based physiological sensing system. We received the Infineon BGT60TR13C mmWave radar and the Polar H10 chest strap, which was used to obtain reference ECG measurements. We collected radar data using the BGT60TR13C and I analyzed the structure and storage format of the acquired radar data to understand how the raw radar measurements are organized and how the different dimensions of the recorded data correspond to the radar acquisition parameters. In parallel, I collected reference ECG data using the Polar H10 for comparison with the radar-based measurements. Based on the initial hardware and data collection, We prepared the materials for the feasibility presentation. I developed the presentation slides covering the research motivation, problem statement, proposed sensing approach, experimental setup, initial data collection, planned system development and project timeline. The slides were reviewed and revised based on feedback from my supervisor to improve the technical clarity and presentation of the proposed work. In addition, we started the hardware design phase of the project and began discussions on the PCB architecture as a group. We evaluated the components required for the wearable prototype and discussed the selection of the optical PPG sensor, the required number of PPG channels, and the overall sensor configuration. These discussions helped us finalize the main components and requirements for the PCB.	
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Index no.: 220327F	Name: Kopithan M.
Detailed list of work carried out (including problems encountered and solutions). During August, I first looked into the wireless signal transmission technologies that could be used to send the recorded data out of the radar device, so that we know what options are available before fixing the rest of the hardware. I then moved from the usual application interface to a more direct way of using the sensor, and configured the radar through its Python wrapper, which allowed us to set the acquisition parameters and collect the raw ADC data generated by the radar instead of GUI. Together with Mathujan, I compared the available radar transceivers and the MAX family PPG sensors and selected the ones that suit our wearable device, and we also studied the datasheets to find the allowed ranges of the operating frequency, samples per chirp, chirp repetition time and frame rate, and the registers that have to be written to control each of them. After that we started the initial PCB planning. I read the sensor datasheet to understand the layout requirements and began selecting the component sizes to be used on the board. The datasheet states that the capacitors placed within about a 5 mm radius of the sensor should be 0402 or 0201 size, and that 0402 or 0603 parts can be used within about a 10 mm radius, so this sets a limit on how small the components near the radar have to be. I also researched the use of an FRAM to store the raw data samples and worked out how much memory would be needed for recording. Finally, we all worked together to collect the first raw radar data, setting the acquisition parameters through the evaluation board application GUI and saving the recorded data so that it can be taken forward for further processing.	
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Index no.: 220389U	Name: Mathujan S.
Detailed list of work carried out (including problems encountered and solutions). During August, I updated the research proposal to match the new direction of the project. Our first draft did not include blood pressure, so I added a section on blood pressure estimation with the relevant citations, and rewrote the project goals after the group redefined them. I then revised the ethics documents and the participant information sheet so that the description given to volunteers matches these redefined goals. Together with Kopithan, I also worked on the comparison and selection of the main hardware components. For the radar transceiver we compared the Texas Instruments IWR1443 and IWR16843 with the Infineon BGT60TR13C, and selected the BGT60TR13C because it is cheaper, has the antennas inside the package, and is small enough for a wrist-worn device. For the optical PPG sensor we compared the MAX86176 and the MAX30101 and selected the MAX86176, since it is built to a clinical standard, is designed for wearable products, and was recommended by a senior who had tested both and found it more sensitive when capturing the PPG signal at the wrist. For the feasibility report I designed the block diagram of the prototype architecture and prepared the budget plan with a summary table of all expected costs, which came to about 644.2 US dollars. I also went through the BGT60TR13C datasheet to find the allowed ranges of the operating frequency, samples per chirp, chirp repetition time and frame rate, and to identify which registers have to be configured to control them, as our adaptive idea depends on changing these settings. Finally, we all worked together to collect the first raw radar data, setting the acquisition parameters through the evaluation board application GUI and saving the recorded data so that it can be taken forward for further processing.	
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Index no.: 220619D	Name: Sivamynthan N.
Detailed list of work carried out (including problems encountered and solutions). This month my work moved from the concept-stage idea of an adaptive radar raised at the end of July into a concrete algorithmic design. I studied the constrained linear contextual bandit formulation used by Thornton et al. for radar parameter selection and adapted it to our setting, using the IMU-derived motion state as the context and a signal-quality-based reward to drive per-CPI (coherent processing interval) reconfiguration of chirp parameters such as gain, chirps per frame and frame rate. Since the reward needed to reflect how well the cardiac pulse could be recovered from a given configuration, I defined a metric I call CSNIR (Cardiac Signal-to-Interference-plus-Noise Ratio), expressed as the log-ratio of spectral power inside the cardiac frequency band to power outside it. I initially considered computing this reward from the phase of the range-FFT bin, since phase is the conventional way small skin displacement is captured in radar literature, but after discussion with the group I decided to keep the reward computation in the amplitude/autocorrelation domain instead, to stay consistent with the rest of our processing pipeline and to be more robust to the kind of motion artefacts we expect during walking and running. While trying to move this from design to implementation, I ran into a hardware limitation. The BGT60TR13C evaluation board runs on a fixed, pre-programmed GUI (Radar Fusion GUI) that does not expose the low-level, per-CPI control needed to reconfigure chirp parameters in real time based on a computed reward. This means the closed-loop adaptive algorithm cannot actually be deployed on the current development board as it stands. After raising this with the group, we agreed that implementing the real-time adaptive loop is not feasible on the eval kit and will instead be built into the custom PCB in a later phase, where we will have direct MCU-level access to the radar's chirp configuration. For now, the eval kit will be used to collect baseline pilot data under fixed settings. Alongside this, I worked on the feasibility report, adding the baseline hardware parameters to the system description and writing up the CSNIR formulation, and helped prepare the feasibility presentation summarising the system architecture and this adaptive design decision for our supervisors. I also continued work on the ethics documentation, including the Sinhala and Tamil translations of the participant information sheet.	
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Supervisor's Certification

Supervisor's Name Dr.Sampath Perera	Signature 	Date
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